

RTL Knowledge Graph Evaluation Report

1. KG Construction Criteria

The KG was built from the full merged ontology seed and merged label set, not from a 100-file sample.

Construction rules:

- Nodes: module, ip_block, label, port
- Edges: HAS_LABEL, HAS_PORT, INSTANTIATES
- Sources: out/merged_ontology_seed.jsonl and out/merged_labels.jsonl
- Projects: OpenTitan and Ibex
- Labels are inferred from module names, paths, ports, and instance clues

2. Full KG Summary

Modules: 1544

IP blocks: 38

Ports: 20196

Instance edges: 14698

Total nodes: 21792

Total edges: 39268

Projects: {'opentitan': 1221, 'ibex': 323}

Top labels: [['hierarchical', 1120], ['clocked', 1067], ['resettable', 1026], ['uart', 341], ['fifo', 268], ['spi', 164], ['crypto', 93], ['controller', 57], ['apb', 48], ['i2c', 32], ['ibex_core', 30], ['counter', 13], ['timer', 13], ['alu', 1]]

3. Baseline QA Benchmark

Question count: 150

Question types: {'exact_name': 50, 'label_plus_ports': 21, 'parent_child': 17, 'path_plus_port': 12, 'reverse_graph': 20, 'multi_hop': 15, 'semantic_bridge': 15}

Project mix: {'ibex': 55, 'opentitan': 95}

4. Baseline Retrieval Performance

baseline: hit@1=0.78, hit@3=0.82, mrr=0.8019, weighted_hit@1=0.7511, proxy_score=78.54

baseline by difficulty: {'easy': {'count': 50, 'hit_at_1': 1.0, 'hit_at_3': 1.0, 'mrr': 1.0}, 'hard': {'count': 50, 'hit_at_1': 0.74, 'hit_at_3': 0.86, 'mrr': 0.7967}, 'medium': {'count': 50, 'hit_at_1': 0.6, 'hit_at_3': 0.6, 'mrr': 0.609}}

kg: hit@1=0.7933, hit@3=0.8333, mrr=0.8169, weighted_hit@1=0.7644, proxy_score=79.9

kg by difficulty: {'easy': {'count': 50, 'hit_at_1': 1.0, 'hit_at_3': 1.0, 'mrr': 1.0}, 'hard': {'count': 50, 'hit_at_1': 0.74, 'hit_at_3': 0.84, 'mrr': 0.7907}, 'medium': {'count': 50, 'hit_at_1': 0.64, 'hit_at_3': 0.66, 'mrr': 0.66}}

Retrieval metadata: {'modules_indexed': 732, 'approved_labels': {'path': 'D:\\MyWork\\verilog\\out\\label_approval\\auto_approved_labels.jsonl', 'module_labels_added': 24, 'ip_context_labels_added': 69}, 'kg_version_tag': None}

5. Multi-Axis Question Set

Total questions: 175

Per level: {'L1': 35, 'L2': 35, 'L3': 35, 'L4': 35, 'L5': 35}

Per type: {'structure_understanding': 25, 'search_navigation': 25, 'comparison_similarity': 25, 'function_similarity': 25, 'generation_design': 25, 'code_explanation': 25, 'documentation_summary': 25}

Matrix counts:

- L1: {'structure_understanding': 5, 'search_navigation': 5, 'comparison_similarity': 5, 'function_similarity': 5, 'generation_design': 5, 'code_explanation': 5, 'documentation_summary': 5}
- L2: {'structure_understanding': 5, 'search_navigation': 5, 'comparison_similarity': 5, 'function_similarity': 5, 'generation_design': 5, 'code_explanation': 5, 'documentation_summary': 5}
- L3: {'structure_understanding': 5, 'search_navigation': 5, 'comparison_similarity': 5, 'function_similarity': 5, 'generation_design': 5, 'code_explanation': 5, 'documentation_summary': 5}
- L4: {'structure_understanding': 5, 'search_navigation': 5, 'comparison_similarity': 5, 'function_similarity': 5, 'generation_design': 5, 'code_explanation': 5, 'documentation_summary': 5}
- L5: {'structure_understanding': 5, 'search_navigation': 5, 'comparison_similarity': 5, 'function_similarity': 5, 'generation_design': 5, 'code_explanation': 5, 'documentation_summary': 5}

6. Multi-Axis Retrieval Performance

baseline: hit@1=0.8514, hit@3=0.8629, mrr=0.8686, weighted_hit@1=0.8393

By level:

L1: hit@1=0.8571, hit@3=0.8857, mrr=0.8714, count=35

L2: hit@1=1.0, hit@3=1.0, mrr=1.0, count=35

L3: hit@1=0.8571, hit@3=0.8571, mrr=0.8857, count=35

L4: hit@1=0.6857, hit@3=0.7143, mrr=0.7286, count=35

L5: hit@1=0.8571, hit@3=0.8571, mrr=0.8571, count=35

By type:

code_explanation: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

comparison_similarity: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

documentation_summary: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

function_similarity: hit@1=0.56, hit@3=0.6, mrr=0.66, count=25

generation_design: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

search_navigation: hit@1=0.6, hit@3=0.6, mrr=0.6, count=25

structure_understanding: hit@1=0.8, hit@3=0.84, mrr=0.82, count=25

kg: hit@1=0.8514, hit@3=0.8629, mrr=0.8562, weighted_hit@1=0.8423

By level:

L1: hit@1=0.8286, hit@3=0.8857, mrr=0.8524, count=35

L2: hit@1=1.0, hit@3=1.0, mrr=1.0, count=35

L3: hit@1=0.8571, hit@3=0.8571, mrr=0.8571, count=35

L4: hit@1=0.7143, hit@3=0.7143, mrr=0.7143, count=35

L5: hit@1=0.8571, hit@3=0.8571, mrr=0.8571, count=35

By type:

code_explanation: hit@1=0.96, hit@3=1.0, mrr=0.9733, count=25

comparison_similarity: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

documentation_summary: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

function_similarity: hit@1=0.56, hit@3=0.6, mrr=0.58, count=25

generation_design: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

search_navigation: hit@1=0.6, hit@3=0.6, mrr=0.6, count=25

structure_understanding: hit@1=0.84, hit@3=0.84, mrr=0.84, count=25

Multi-axis metadata: {'modules_indexed': 732, 'approved_labels': {'path':

'D:\\MyWork\\verilog\\out\\label_approval\\auto_approved_labels.jsonl', 'module_labels_added':

24, 'ip_context_labels_added': 69}, 'kg_version_tag': None}

7. Manticore Search Retrieval

This evaluation adds a Manticore Search-style BM25F full-text ranker over parser/LSP fields.

It also reports a hybrid variant that indexes KG labels, summaries, and reverse parent context.

manticore_parser_lsp: hit@1=0.8629, hit@3=0.8914, mrr=0.8857, weighted_hit@1=0.8401

manticore_hybrid: hit@1=0.8686, hit@3=0.8914, mrr=0.8781, weighted_hit@1=0.8479

Manticore metadata: {'modules_indexed': 732, 'questions': 175, 'seed':

'D:\\MyWork\\verilog\\out\\merged_ontology_seed.jsonl', 'approved_labels': {'path':

'D:\\MyWork\\verilog\\out\\label_approval\\auto_approved_labels.jsonl', 'module_labels_added':

24, 'ip_context_labels_added': 69}, 'manticore_repo':

'D:\\MyWork\\verilog\\tools\\manticoresearch', 'manticore_source':

'https://github.com/manticoresoftware/manticoresearch', 'index_build_ms': 272.225,

'latency_ms': {'baseline': {'total_ms': 4929.941, 'avg_query_ms': 28.171}, 'kg': {'total_ms':

5999.671, 'avg_query_ms': 34.284}, 'manticore_parser_lsp': {'total_ms': 6542.472,

'avg_query_ms': 37.386}, 'manticore_hybrid': {'total_ms': 8123.22, 'avg_query_ms': 46.418}},

'note': 'Local BM25F model of Manticore Search over parser+LSP fields; schema/documents are emitted for real server loading.'}

8. VerilogEval Status

Official verilogeval runner/package was not available in this workspace.

All score_100 numbers are proxy scores, not official VerilogEval scores.